

FULL TITLE OF THE PAPER

AUTHOR NAME

ABSTRACT. Write the abstract here. Summarize the main results and techniques.

1. INTRODUCTION

Write the introduction here. Use inline math like $f: X \rightarrow Y$.
A single numbered equation:

$$(1.1) \quad \int_0^1 f(x) \, dx = F(1) - F(0).$$

Reference it with (1.1).

2. MAIN RESULTS

Theorem 2.1 (Main Theorem). *Let X be a complete metric space and $T: X \rightarrow X$ a contraction mapping with constant $k < 1$. Then T has a unique fixed point $x^* \in X$.*

Proof. Define the sequence $x_{n+1} = T(x_n)$ for some starting point $x_0 \in X$. We show $\{x_n\}$ is Cauchy:

$$\begin{aligned} \|x_{n+1} - x_n\| &= \|T(x_n) - T(x_{n-1})\| \\ &\leq k\|x_n - x_{n-1}\| \\ &\leq k^n\|x_1 - x_0\|. \end{aligned}$$

Since $k < 1$, the geometric series converges, so $\{x_n\}$ is Cauchy. Completeness gives convergence; continuity of T gives the fixed point. $\square$

Corollary 2.2. *The fixed point depends continuously on the contraction map T .*

Definition 2.3. A function $f: X \rightarrow Y$ between metric spaces is *Lipschitz* if there exists $L \geq 0$ such that $\|f(x) - f(y)\| \leq L\|x - y\|$ for all $x, y \in X$.

Remark 2.4. Every contraction is Lipschitz with constant $L = k < 1$.

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Key words and phrases. keyword one, keyword two, keyword three.

$$\begin{aligned}
 (2.1) \quad & a = b + c \\
 & = d + e + f \\
 & = g.
 \end{aligned}$$

$$(2.2) \quad \alpha = \beta + \gamma,$$

$$(2.3) \quad \delta = \epsilon.$$

3. CONCLUSION

Summarize results and discuss future directions.

DEPARTMENT OF MATHEMATICS, UNIVERSITY, CITY, COUNTRY
Email address: `author@university.edu`